DIMENSION IS CONSTANT ON NONEMPTY OPENS OF IRREDUCIBLE JACOBSON CATENARY SCHEMES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-geometry.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for stacks-geometry.tex (ge-occupied.json, 2026-10-14); 207 route=geometry / chapter-list field hits in focused-seeds are alias noise (route name and alphabetical chapter enumerations), excluded per the ladder alias-blind-spot caveat.

Source reference. Original reference 081I: sources/stacks-chow/tags.txt, source lines 10218–10218. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02IM: sources/stacks-chow/tags.txt, source lines 3305–3305. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *If X is an irreducible, Jacobson, catenary, and locally Noetherian scheme of finite dimension, then $\dim U = \dim X$ for every non-empty open subset U of X. Equivalently, $\dim_x X$ is a constant function on X.*

Proof. The equivalence of the two claims follows directly from the definitions. Suppose, then, that $U \subset X$ is a non-empty open subset. Certainly $\dim U \leq \dim X$, and we have to show that $\dim U \geq \dim X$. Write $d = \dim X$, and choose a maximal strictly decreasing chain of irreducible closed subsets of X, say

$$X = Z_0 \supset Z_1 \supset \ldots \supset Z_d.$$

Since X is Jacobson, the minimal irreducible closed subset Z_d is equal to $\{x\}$ for some closed point x.

If $x \in U$, then

$$U = U \cap Z_0 \supset U \cap Z_1 \supset \ldots \supset U \cap Z_d = \{x\}$$

is a strictly decreasing chain of irreducible closed subsets of U, and so we conclude that $\dim U \geq d$, as required. Thus we may suppose that $x \notin U$.

Consider the flat morphism $\operatorname{Spec} \mathcal{O}_{X,x} \to X$. The non-empty (and hence dense) open subset U of X pulls back to an open subset $V \subset \operatorname{Spec} \mathcal{O}_{X,x}$. Replacing U by a non-empty quasi-compact, and hence Noetherian, open subset, we may assume that the inclusion $U \to X$ is a quasi-compact morphism. Since the formation

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of scheme-theoretic images of quasi-compact morphisms commutes with flat base-change Morphisms, Lemma 081I we see that V is dense in $\operatorname{Spec}\mathcal{O}_{X,x}$, and so in particular non-empty, and of course $x \notin V$. (Here we use x also to denote the closed point of $\operatorname{Spec}\mathcal{O}_{X,x}$, since its image is equal to the given point $x \in X$.) Now $\operatorname{Spec}\mathcal{O}_{X,x} \setminus \{x\}$ is Jacobson Properties, Lemma 02IM and hence V contains a closed point z of $\operatorname{Spec}\mathcal{O}_{X,x} \setminus \{x\}$. The closure in X of the image of z is then an irreducible closed subset Z of X containing x, whose intersection with U is non-empty, and for which there is no irreducible closed subset properly contained in Z and properly containing $\{x\}$ (because pull-back to $\operatorname{Spec}\mathcal{O}_{X,x}$ induces a bijection between irreducible closed subsets of X containing x and irreducible closed subsets of $\operatorname{Spec}\mathcal{O}_{X,x}$). Since $U \cap Z$ is a non-empty closed subset of U, it contains a point u that is closed in X (since X is Jacobson), and since $U \cap Z$ is a non-empty (and hence dense) open subset of the irreducible set Z (which contains a point not lying in U, namely x), the inclusion $\{u\} \subset U \cap Z$ is proper.

As X is catenary, the chain

$$X = Z_0 \supset Z \supset \{x\} = Z_d$$

can be refined to a chain of length $d+1$, which must then be of the form

$$X = Z_0 \supset W_1 \supset \ldots \supset W_{d-1} = Z \supset \{x\} = Z_d.$$

Since $U \cap Z$ is non-empty, we then find that

$$U = U \cap Z_0 \supset U \cap W_1 \supset \ldots \supset U \cap W_{d-1} = U \cap Z \supset \{u\}$$

is a strictly decreasing chain of irreducible closed subsets of U of length $d+1$, showing that $\dim U \geq d$, as required. $\square$